# Consumption Decision Analysis Report

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## Executive Summary

This project analyzes primary survey data (n=30) collected through a structured questionnaire among young adults.

After cleaning and transforming categorical spending ranges into numeric midpoint values, a multiple linear regression model was estimated.

The regression results show that demographic variables explain only a small portion of spending variation ($R^2 = 0.11$), indicating limited predictive power.

The findings suggest that demographic segmentation alone may be insufficient to predict consumption behavior in this group. Behavioral and income-related variables may provide stronger explanatory power.

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## 1. Business Motivation

Understanding consumption patterns among young adults is important for businesses targeting student and early-career markets. This study investigates whether demographic factors—specifically location, student status, and age group—are associated with differences in monthly personal spending.

The goal is to explore whether these characteristics meaningfully explain spending variation and to evaluate the strength of demographic segmentation in predicting consumption levels.

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## 2. Data Description

The dataset consists of 30 survey responses collected through a structured questionnaire.

Data Cleaning and Transformation

Monthly spending was originally recorded as categorical ranges (e.g., “2,000–4,000 RMB”). To enable quantitative analysis, each range was converted into a numeric value using the midpoint of the interval.

Categorical predictors (location, student status, age group) were encoded as dummy variables in R prior to model estimation.

No missing values were found after cleaning.

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## 3. Descriptive Statistics

The sample consists of:

- 10 respondents from Mainland China
- 5 from Other Countries
- 15 from the United States
- 28 out of 30 respondents are students
- Majority are aged 18–22

Because of the small and uneven distribution across categories, potential statistical power limitations are expected.

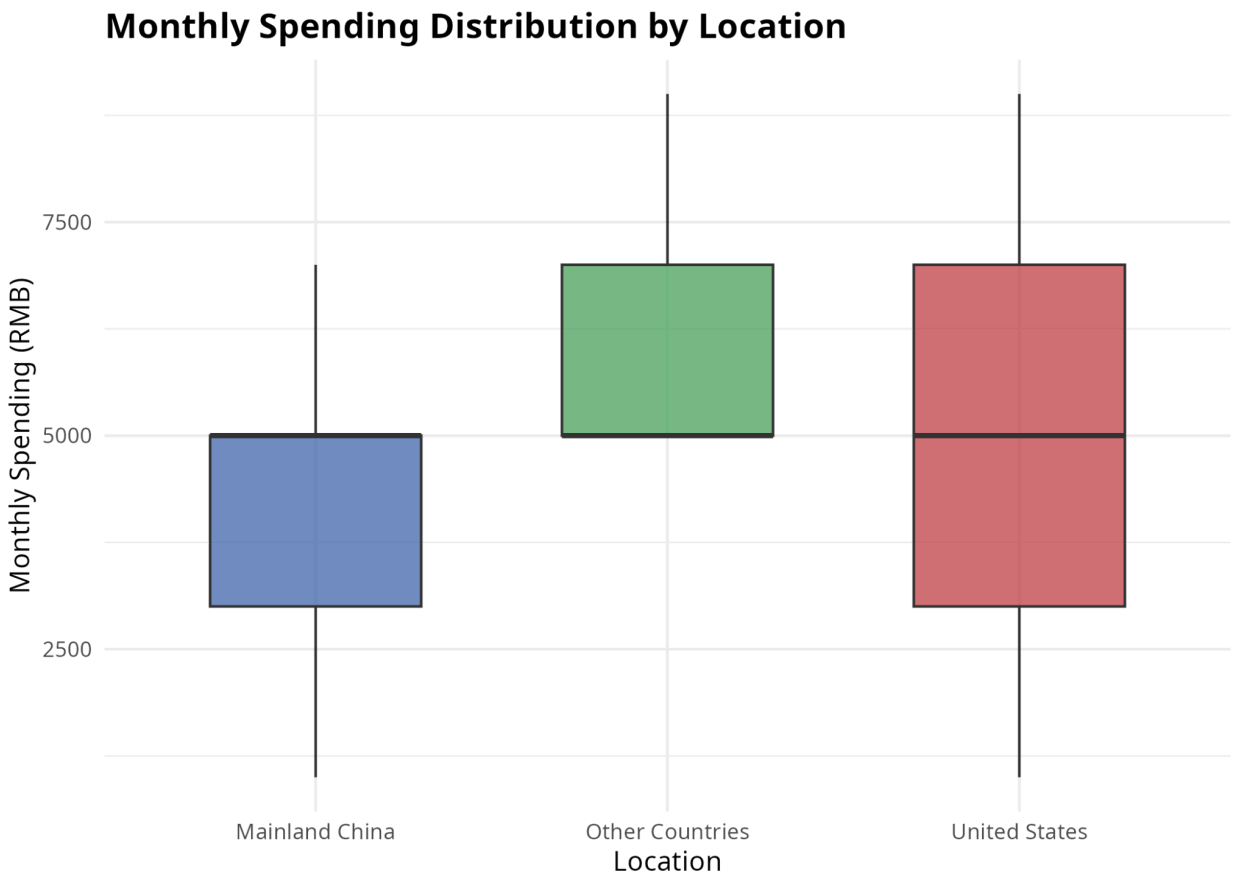


**Figure 1**. Monthly Spending Distribution by Location.

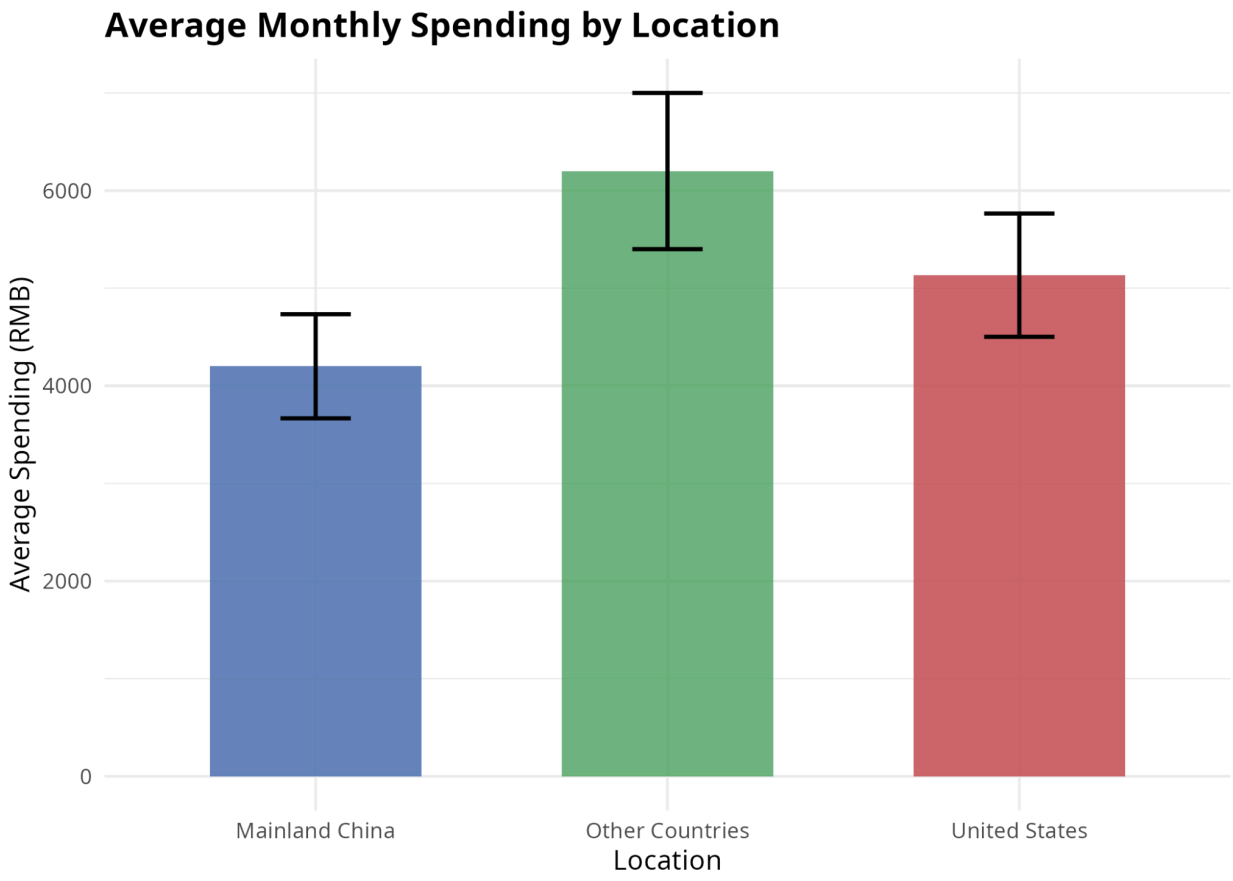


**Figure 2** shows the group means with standard error bars, indicating that differences across locations are not clearly separated.

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## 4. Model Specification

We estimated a multiple linear regression model:

spend_num ~ location + student + age

The sample is dominated by students aged 18–22, which limits variation and may reduce the model's ability to detect demographic effects.

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## 5. Regression Results

The overall model was not statistically significant:

- $F(5, 24) = 0.60$, $p = 0.70$

- $R^2 = 0.11$

Only 11% of the variance in spending is explained by the included demographic predictors.

The estimated coefficients were relatively small, and their large standard errors suggest substantial uncertainty in the estimates. Individual coefficients for location, student status, and age group were not statistically significant at the 5% level, suggesting that no single demographic factor shows a strong independent effect on spending within this sample.

Mainland China, non-students, and the 18–22 age group were used as reference categories.

Given the small sample and limited variation, the lack of statistical significance should be interpreted with caution rather than taken as definitive evidence of no effect.

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## 6. Robustness & Model Diagnostics

To future verify the regression result, I conducted a diagnostic test on this. First, I performed a logarithmic transformation on the expenditure data and re-estimated another model, but the result still lacked statistical significance. This suggests that the weak explanatory power is unlikely to be driven by distributional issues in the spending variable.

Second, I examined multicollinearity by calculating the Variance Factors (VIF). All VIF values were near 1, which suggests that there was no significant multicollinearity between the independent variables.

Finally, I checked the residual plots and found no major violations of the assumptions of linearly, normally, and homoscedasticity. Using leverage values and Cook's distance, I also found that no observations had an overly strong influence on the result.

Overall, the explanatory power of this model is not strong, mainly due to the small sample size and the lack of some key behavior-related variables.

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## 7. Interpretation & Business Implications

Within this sample, demographic factors alone do not strongly predict spending differences among young adults.

Possible explanations:

1. Spending behavior may be more influenced by income level, lifestyle, or consumption preferences rather than basic demographics.
2. The sample is highly unbalanced (93% students), limiting variability.
3. The small sample size (n=30) reduces statistical power.
4. Converting spending ranges into midpoint values introduces measurement noise.

From a business perspective, these findings indicate that relying solely on demographic segmentation may not effectively differentiate spending behavior among young adults. Companies targeting this market may benefit from incorporating behavioral data, income level, lifestyle indicators, or consumption preferences to improve predictive accuracy and customer segmentation strategies.

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## 8. Limitations & Future Improvements

Limitations:

- Small sample size (n=30)
- Uneven group distribution (93% students)
- Spending measured using interval midpoints (approximation error)

Future improvements:

- Increase sample size to at least 150–200 observations
- Collect continuous income data instead of spending ranges
- Incorporate behavioral variables (e.g., spending categories, lifestyle habits)
- Compare alternative modeling approaches (e.g., log transformation, robust regression)